ASSIGNMENT-9

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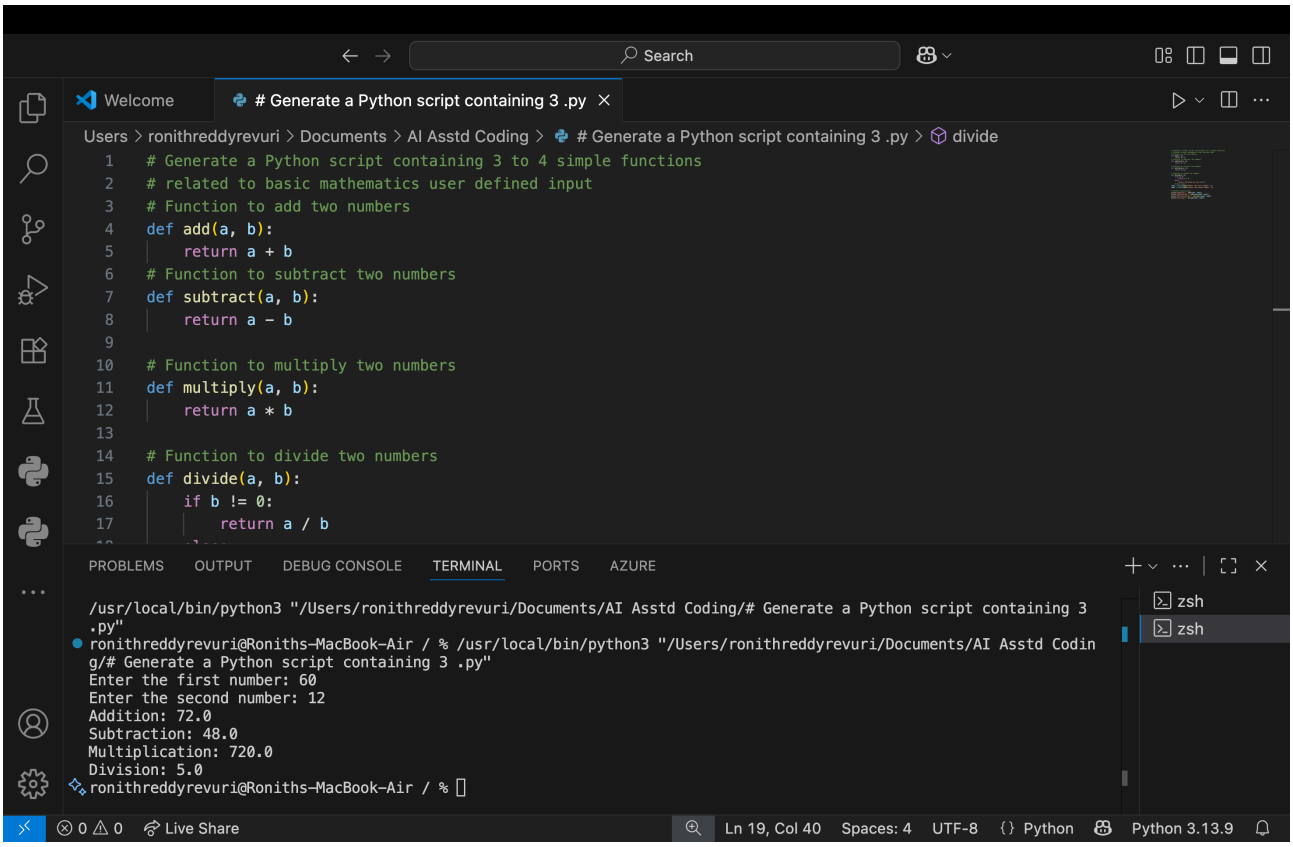
TASK DESCRIPTION -1 (DOCUMENTATION – FUNCTION SUMMARY GENERATION)

PROMPT USED :
 # GENERATE A PYTHON SCRIPT CONTAINING 3 TO 4 SIMPLE FUNCTIONS
 # RELATED TO BASIC MATHEMATICS USER DEFINED INPUT

EXPLANATION :

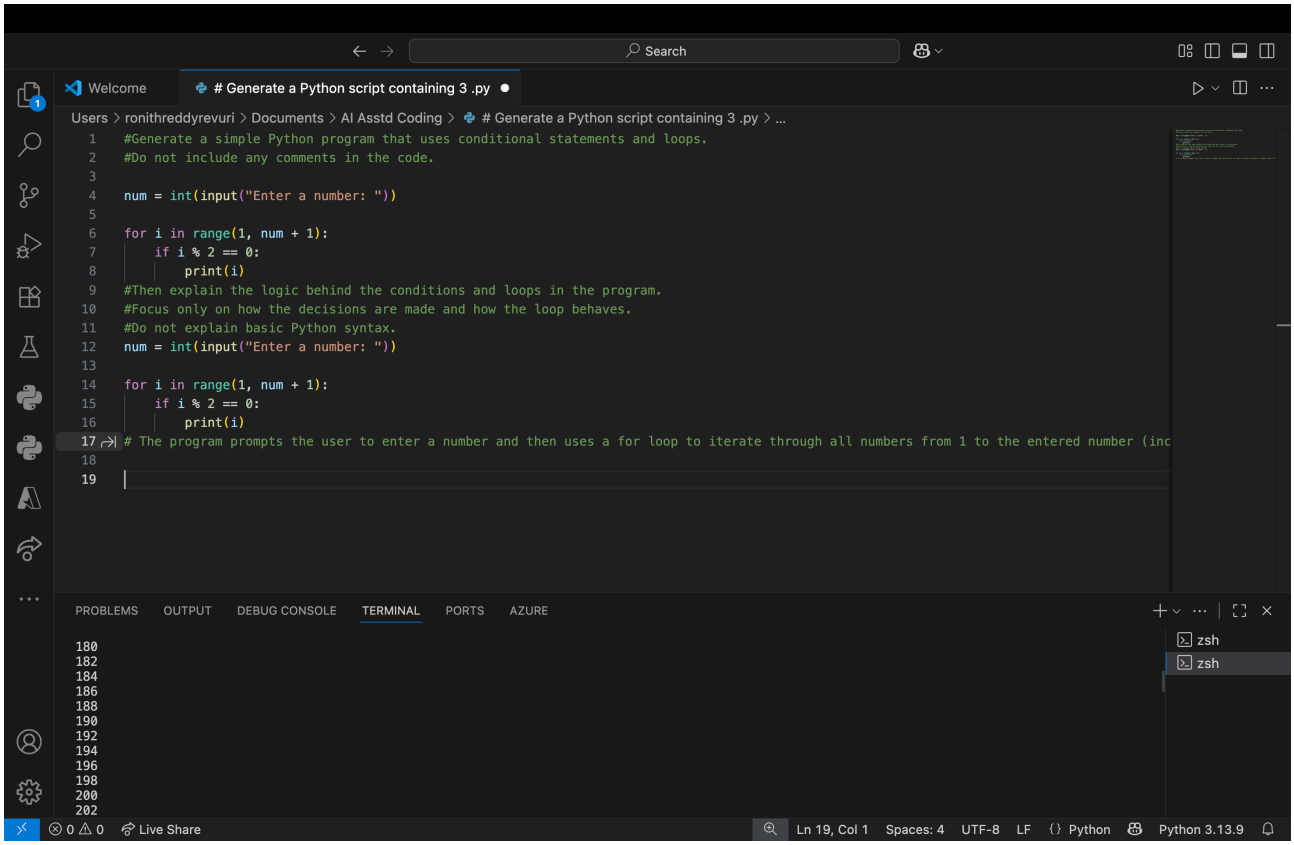
FIRST, A PYTHON SCRIPT WITH FUNCTIONS IS CREATED USING AI. THEN, SUMMARIES FOR EACH FUNCTION ARE CREATED USING AI.
THIS DEMONSTRATES HOW AI CAN BE USED FOR AUTOMATIC DOCUMENTATION GENERATION.



TASK -2 : TASK DESCRIPTION -2 (DOCUMENTATION – LOGICAL EXPLANATION FOR CONDITIONS AND LOOPS)

PROMPT USED :
GENERATE A SIMPLE PYTHON PROGRAM THAT USES CONDITIONAL STATEMENTS AND LOOPS.
DO NOT INCLUDE ANY COMMENTS IN THE CODE.

THEN EXPLAIN THE LOGIC BEHIND THE CONDITIONS AND LOOPS IN THE PROGRAM.
FOCUS ONLY ON HOW THE DECISIONS ARE MADE AND HOW THE LOOP BEHAVES.
DO NOT EXPLAIN BASIC PYTHON SYNTAX.

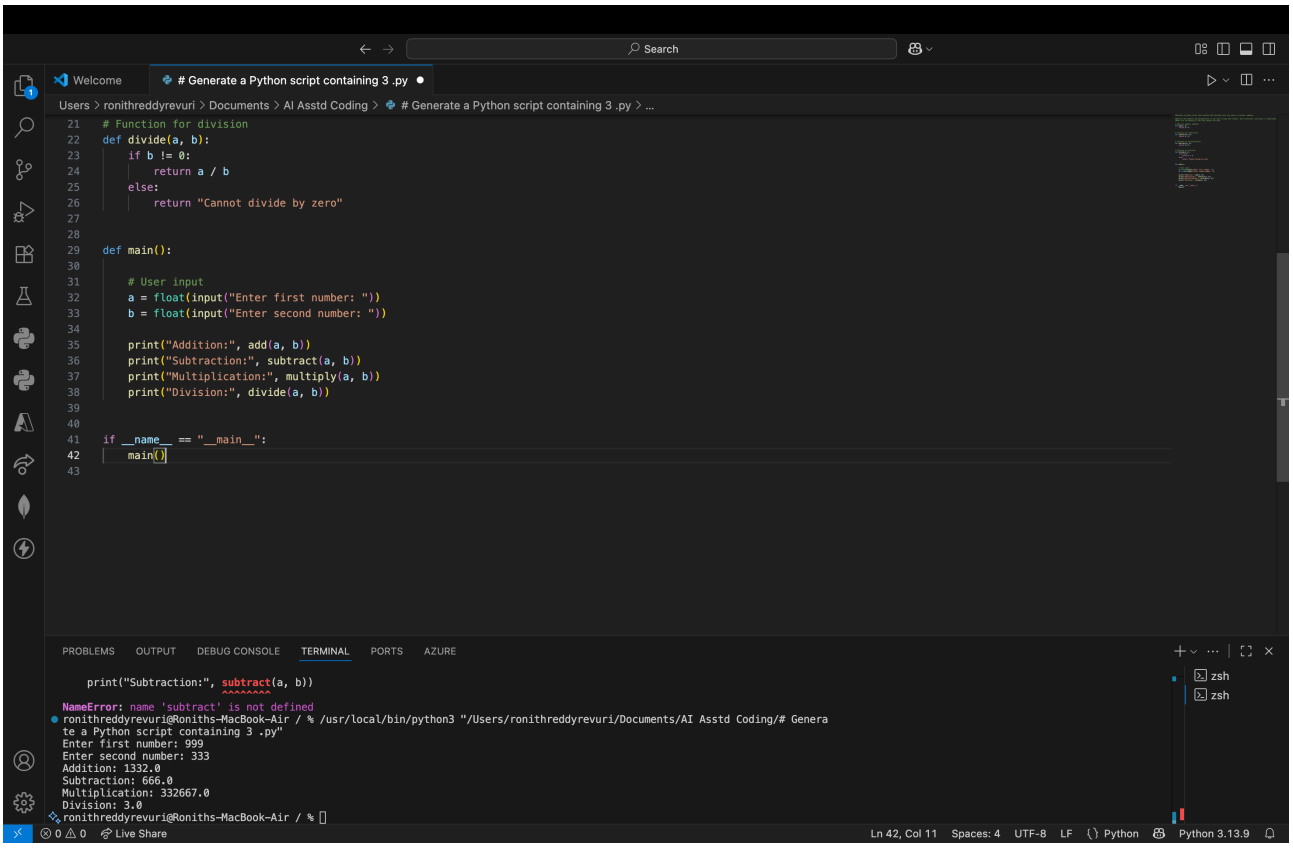


EXPLANATION :

IN THIS TASK, AI GENERATES A PYTHON PROGRAM THAT INCLUDES A LOOP AND A CONDITIONAL STATEMENT.
THE LOOP RUNS FROM 1 UP TO THE NUMBER ENTERED BY THE USER.
DURING EACH ITERATION, THE PROGRAM CHECKS A CONDITION TO DETERMINE WHETHER THE CURRENT NUMBER IS DIVISIBLE BY 2.

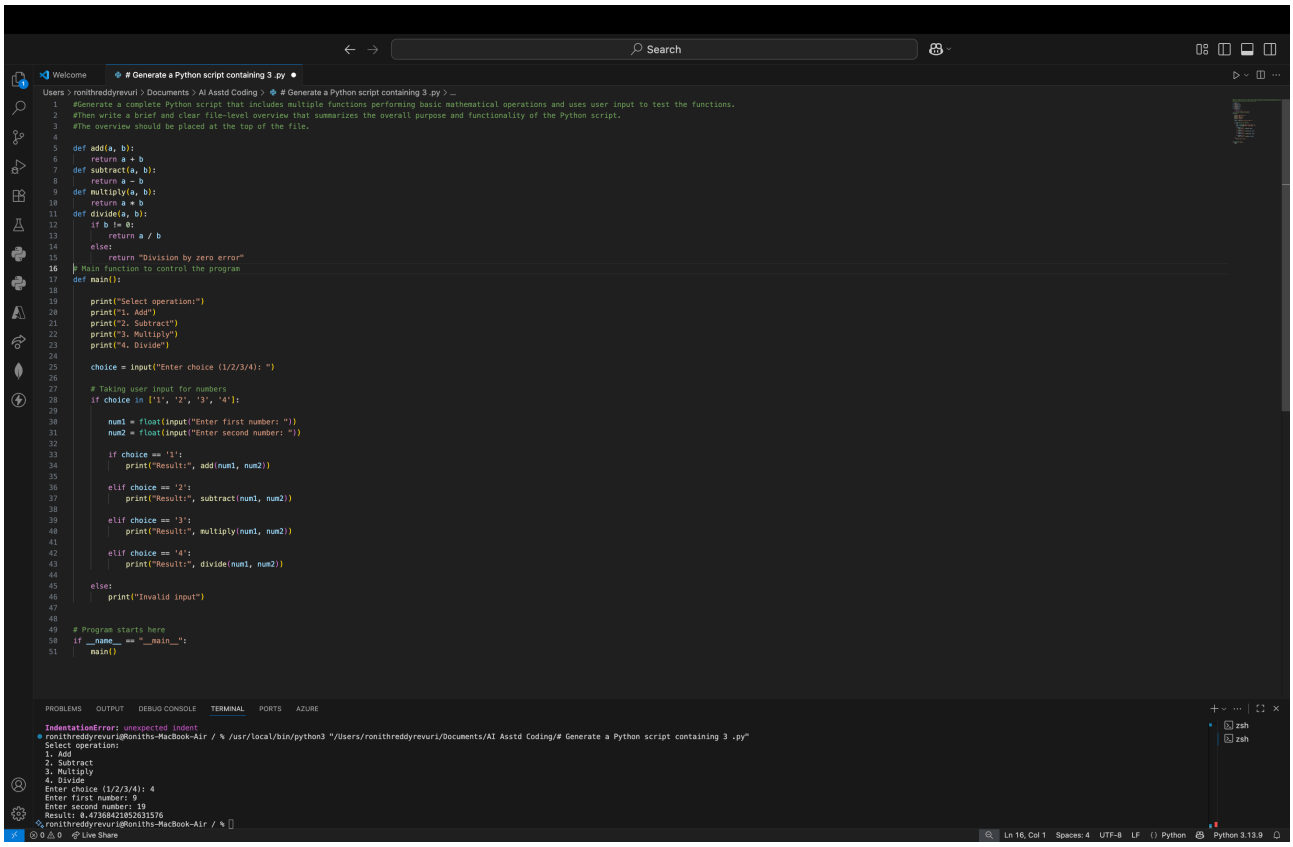
TASK - 3 : DOCUMENTATION: FILE-LEVEL OVERVIEW

PROMPT USED :
GENERATE A COMPLETE PYTHON SCRIPT THAT INCLUDES MULTIPLE FUNCTIONS PERFORMING BASIC MATHEMATICAL OPERATIONS AND USES USER INPUT TO TEST THE FUNCTIONS.



EXPLANATION :
IN THIS TASK, AI FIRST GENERATES A PYTHON SCRIPT THAT CONTAINS MULTIPLE FUNCTIONS FOR PERFORMING MATHEMATICAL OPERATIONS.
AFTER THAT, A FILE-LEVEL OVERVIEW IS GENERATED TO DESCRIBE THE OVERALL PURPOSE OF THE PROGRAM.

TASK- 4 :
DOCUMENTATION: REFINE EXISTING DOCUMENTATION



PROMPT :
GENERATE A PYTHON SCRIPT THAT CONTAINS 3–4 FUNCTIONS WITH VERY BASIC OR UNCLEAR COMMENTS.

REWRITE THE COMMENTS AND DOCUMENTATION IN THE CODE TO MAKE THEM CLEARER, MORE CONSISTENT, AND EASIER TO UNDERSTAND.
MAKE SURE THE MEANING OF THE CODE REMAINS THE SAME.

EXPLANATION :

IN THIS TASK, AI GENERATES PYTHON CODE THAT INITIALLY CONTAINS VERY BASIC OR UNCLEAR COMMENTS.
THEN AI IMPROVES THESE COMMENTS BY REWRITING THEM IN A CLEARER AND MORE CONSISTENT WAY.
THE LOGIC OF THE PROGRAM REMAINS UNCHANGED, BUT THE DOCUMENTATION BECOMES EASIER TO READ AND UNDERSTAND.
THIS PROCESS HELPS DEVELOPERS IMPROVE CODE DOCUMENTATION QUALITY WITHOUT MODIFYING THE ACTUAL PROGRAM LOGIC.

TASK - 5 : DOCUMENTATION: PROMPT DETAIL IMPACT STUDY

PROMPT:

GENERATE A SIMPLE PYTHON FUNCTION THAT PERFORMS A REAL-WORLD CALCULATION.

THEN WRITE A CLEAR AND COMPLETE DOCSTRING EXPLAINING:
- THE PURPOSE OF THE FUNCTION
- THE INPUT PARAMETERS
- THE RETURN VALUE

EXPLANATION :
IN THIS TASK, A SIMPLE PYTHON FUNCTION IS CREATED TO CALCULATE THE FINAL PRICE AFTER APPLYING A DISCOUNT.
THE FUNCTION TAKES TWO INPUTS: THE ORIGINAL PRICE OF THE PRODUCT AND THE DISCOUNT PERCENTAGE.
THE FUNCTION THEN CALCULATES THE DISCOUNT AMOUNT AND SUBTRACTS IT FROM THE ORIGINAL PRICE TO PRODUCE THE FINAL PRICE.

Search

RUN AND DEBUG: ...

Run and Debug

To customize Run and Debug, open a folder and create a launch.json file.

Debug using a terminal command or in an interactive chat.

Show automatic Python configurations

Welcome | # Generate a Python script containing 3 .py

Users > ronithreddyrevuri > Documents > AI Asstd Coding > # Generate a Python script containing 3 .py > ...

```python
#Generate a simple Python function that performs a real-world calculation.
#Then create two different prompts:
#1. A short prompt
#2. A detailed prompt
#Use both prompts to generate documentation for the same function and compare the results.

def calculate_discount(price, discount):
    return price - (price * discount / 100)
def calculate_discount(price, discount):
    """
    Calculate the discounted price based on the original price and discount percentage.

    Parameters:
    price (float): The original price of the item.
    discount (float): The discount percentage to be applied.

    Returns:
    float: The final price after applying the discount.
    """
    return price - (price * discount / 100)
# Taking input from the user
price = float(input("Enter the original price: "))
discount = float(input("Enter the discount percentage: "))

# Calling the function
final_price = calculate_discount(price, discount)

# Displaying the result
print("Final price after discount:", final_price)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

```
ronithreddyrevuri@Roniths-MacBook-Air / % /usr/local/bin/python3 "/Users/ronithreddyrevuri/Documents/AI Asstd Coding/# Genera
te a Python script containing 3 .py"
ronithreddyrevuri@Roniths-MacBook-Air / % /usr/local/bin/python3 "/Users/ronithreddyrevuri/Documents/AI Asstd Coding/# Genera
te a Python script containing 3 .py"
ronithreddyrevuri@Roniths-MacBook-Air / % k
zsh: command not found: k
ronithreddyrevuri@Roniths-MacBook-Air / % /usr/local/bin/python3 "/Users/ronithreddyrevuri/Documents/
AI Asstd Coding/# Generate a Python script containing 3 .py"
Enter the original price: 9000
Enter the discount percentage: 99
Final price after discount: 90.0
ronithreddyrevuri@Roniths-MacBook-Air / %
```

zsh
zsh

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